

SOCRA

Project Vision, AI Architecture & Cost Strategy

Pilot Research V1 → AI-native learning platform

Purpose: define the idea behind Socra, its long-term vision, the AI capabilities required to ship the university pilot, and the most cost-efficient model choices for each capability.

Pricing snapshot: October 6, 2026

All model prices are public list prices in USD per 1 million tokens unless noted.

Planning estimates are not vendor quotes and should be re-checked before production deployment.

1. Executive Summary

Socra is an AI-native learning environment for introductory university computer science. Its core idea is not merely to put a chatbot beside an assignment. Socra changes how AI behaves according to the educational context. During protected homework and quizzes, AI should understand the student's code and reasoning while withholding solution-bearing assistance. During practice and post-assessment review, AI can become a much more direct tutor. The resulting interactions create structured learning signals that help students practice weak areas and help faculty see where the class is struggling.

Strategic recommendation: do not use one expensive model for everything. Use a model router. Reserve a balanced/high-quality model for protected Socratic conversations, and use a very cheap model for cards, tagging, quiz metadata, misconception extraction, analytics preprocessing, and other high-volume structured work.

Recommended Pilot AI Stack

Capability	Recommended V1 model	Why	Typical list price
Protected Socratic tutor	GPT-6.1 Sol	Strong coding/reasoning; controlled quality; same OpenAI stack	\$2 in / \$10 out per 1M
Cheap structured AI / router	GPT-6 Luna	Extremely low unit cost; 1.05M context; structured tasks	\$0.10 in / \$0.50 out
Faculty authoring	GPT-6.1 Sol	Higher quality matters; usage volume is low	\$2 in / \$10 out
Practice question generation	GPT-6 Luna first; Sol fallback	Generate cheaply, escalate hard items	\$0.10/\$0.50 → \$2/\$10
Misconception extraction	GPT-6 Luna	Classification/extraction, not open-ended frontier reasoning	\$0.10 in / \$0.50 out
Weekly teaching brief	GPT-6 Luna or Sol	Numbers should be computed in SQL; model summarizes only	\$0.10/\$0.50 or \$2/\$10
Embeddings / course RAG	text-embedding-3-small	Very low cost; sufficient V1 retrieval baseline	\$0.02 per 1M input tokens

A single-provider OpenAI stack is recommended for the pilot because the team is already using an OpenAI API key. This minimizes engineering variance, credential management, provider-specific failure modes, and research reproducibility problems. Anthropic and Google should remain benchmark/fallback candidates rather than simultaneous production dependencies in the first study.

2. The Idea Behind Socra

General-purpose AI can solve many introductory CS assignments outright. That creates a product paradox: students benefit from immediate help, but the easiest implementation of AI assistance can replace the thinking the assignment was designed to develop. Socra treats this as the central design problem.

2.1 The closed learning loop

- Faculty creates an assignment with explicit learning objectives and course context.
- Student works in a native coding or written-response workspace.
- Socra sees the prompt, current work, errors, output, and approved course resources automatically.
- Protected Socra gives diagnostic and Socratic help without completing the core assessed task.
- The interaction creates structured evidence: attempts, hint depth, misconceptions, recovery, correctness, and topic associations.

- The learner profile updates with evidence-based topic states such as Needs reinforcement, Developing, and Consistently demonstrated.
- Faculty analytics aggregate those signals into class pain points and question-level insight.
- Faculty changes instruction or creates targeted follow-up practice.

2.2 What makes Socra different

Traditional LMS	Generic AI chatbot	Socra
Distributes assignments	Answers questions	Understands assignment + live workspace
Tracks grades	Optimizes for helpful answers	Optimizes for learning context
Little insight into reasoning	May directly solve assessed task	Controls solution-bearing disclosure
Static course flow	Conversation-centric	Assessment + practice + learner model + analytics loop
Faculty sees outputs	No native class intelligence	Converts interactions into aggregate instructional signals

3. Vision and Goals

3.1 Research V1 goal

Ship a complete, reliable, end-to-end system that CSE 115/116 students can actually use during the university pilot. The pilot is not a prototype. It must include authentication, course/assignment state, coding or written workspaces, AI orchestration, secure code execution, persistence, submission, grading, learner data, analytics, research telemetry, privacy controls, observability, and administration.

3.2 Research goal

Test whether constrained Socratic AI can improve learning outcomes, reduce direct answer dependence, preserve cognitive effort, improve retention, and expose useful class-level misconceptions compared with unrestricted AI or conventional support conditions, subject to the approved study design.

3.3 Long-term product goal

Build an AI-native LMS where AI is a context-aware learning layer rather than a detachable chatbot. The long-term platform can expand from introductory CS into STEM, while eventually supporting broader LMS primitives such as courses, rosters, gradebook, submissions, announcements, content modules, and integrations.

3.4 Competitive implication of Canvas + OpenAI

Instructure and OpenAI's partnership validates the category: AI will increasingly be embedded in LMS assignments and learning experiences. That makes 'AI inside an LMS' insufficient differentiation. Socra should concentrate its research and product identity around protected cognitive assistance, CS-native workspace understanding, intervention depth, misconception analytics, and privacy-preserving instructional intelligence.

4. The Different AIs Socra Actually Needs

AI role	Volume	Quality need	Latency need	Structured?	Recommended tier
Protected Socratic tutor	Very high	Very high	High	Mixed	Balanced
Practice tutor	High	High	High	Mixed	Economy/Balanced
Quiz/question	Medium	Medium-High	Low	Yes	Economy

generator					
Interactive card generator	High	Medium	High	Yes	Economy
Misconception extractor	Very high	Medium	Medium	Yes	Economy
Topic/tag classifier	Very high	Medium	Medium	Yes	Economy
Faculty assignment copilot	Low	Very high	Medium	Mixed	Balanced/Premium
Rubric/feedback suggester	Medium	High	Medium	Yes	Balanced
Analytics narrative writer	Low	Medium	Low	Yes	Economy
Policy/leakage checker	High	High	Medium	Yes	Economy/Balanced
Course-material embeddings	Batch	Medium	Low	Vector	Embedding model

4.1 Protected Socratic Tutor

This is the product-critical model. It needs strong coding ability, nuanced instruction following, reliable context use, and the ability to diagnose a student's mistake without immediately solving the assignment. This is where paying for a balanced model is justified. The research study is also cleaner if this student-facing model remains stable for a study cohort rather than dynamically changing between several vendors.

4.2 Cheap structured intelligence

A large fraction of Socra's AI work does not require frontier intelligence. Topic tagging, extracting a misconception label, generating a compact UI card, converting a response into JSON, summarizing computed analytics, or deciding which practice topic to select are high-volume but bounded tasks. These should be routed to the cheapest model that passes an internal accuracy threshold.

4.3 Faculty authoring AI

Faculty authoring is low-volume and high-value. A professor may generate only tens of assignment components, rubrics, and scaffolds per course. Even a 10x more expensive model can be economical here because the total call volume is tiny relative to student chat. Quality should dominate cost for authoring.

4.4 Analytics AI

The model should never invent class statistics. Counts, percentages, gains, and funnels should be computed deterministically in SQL/analytics jobs. AI receives the already-computed metrics and turns them into concise instructional summaries or recommendations.

4.5 Embeddings / RAG

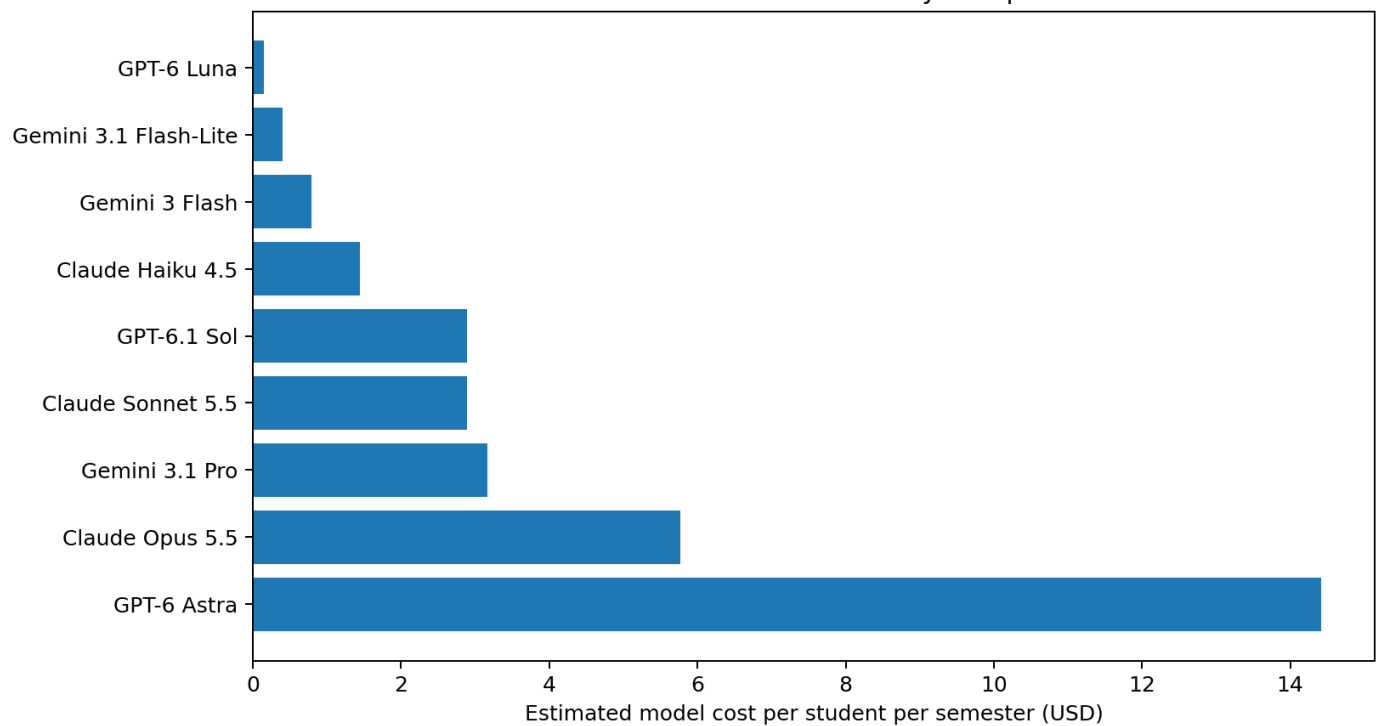
Course resources should be embedded once, chunked, and retrieved by semantic similarity. Embedding cost is typically negligible compared with generative calls. For example, embedding 5 million tokens with OpenAI text-embedding-3-small at \$0.02 per million tokens is approximately \$0.10.

5. Current Model Price Landscape

The table below sorts candidate models by estimated cost for a standardized Socra semester workload, not by benchmark quality. Prices are public list prices observed on October 6, 2026. Preview models, promotional pricing, regional processing, caching, batch processing, reasoning-token behavior, and provider policy can change.

Rank	Provider / model	Input \$/1M	Output \$/1M	3K in + 500 out	Base \$/student	Tier
1	OpenAI — GPT-6 Luna	\$0.10	\$0.50	\$0.00055	\$0.14	Economy
2	Cohere — Command R	\$0.15	\$0.60	\$0.00075	\$0.20	Economy
3	Google — Gemini 3.1 Flash-Lite	\$0.25	\$1.50	\$0.00150	\$0.40	Economy
4	Google — Gemini 3 Flash	\$0.50	\$3.00	\$0.00300	\$0.79	Value
5	Anthropic — Claude Haiku 4.5	\$1.00	\$5.00	\$0.00550	\$1.44	Value
6	OpenAI — GPT-6.1 Sol	\$2.00	\$10.00	\$0.01100	\$2.88	Balanced
7	Anthropic — Claude Sonnet 5.5	\$2.00	\$10.00	\$0.01100	\$2.88	Balanced
8	Google — Gemini 3.1 Pro	\$2.00	\$12.00	\$0.01200	\$3.16	Balanced
9	OpenAI — GPT-5.6 Terra	\$2.00	\$12.00	\$0.01200	\$3.16	Balanced
10	Anthropic — Claude Opus 5.5	\$4.00	\$20.00	\$0.02200	\$5.77	Premium
11	OpenAI — GPT-5.6 Sol	\$4.00	\$20.00	\$0.02200	\$5.77	Premium
12	OpenAI — GPT-6 Astra	\$10.00	\$50.00	\$0.05500	\$14.42	Frontier

Base workload scenario: model-only cost per student



Base workload cost is deliberately an estimate. It assumes 654,000 input tokens and 120,000 output tokens per student over a semester, then adds 15% overhead for retries, metadata, and operational variance. The workload is a planning model, not measured pilot usage.

6. Cost Model Assumptions

Component	Base assumption per student / semester	Tokens
Protected assignment AI	28 sessions × 6 turns × ~3,000 input + 500 output	504K input / 84K output

Practice AI	12 sessions × 5 turns × ~2,500 input + 600 output	150K input / 36K output
Subtotal	Interactive student workload	654K input / 120K output
Operational overhead	15% cost uplift	Retries, guardrails, summaries, variance

6.1 Sensitivity: low / base / high usage

Model	Low usage	Base usage	High usage
GPT-6 Luna	\$0.06	\$0.14	\$0.35
Gemini 3.1 Flash-Lite	\$0.17	\$0.40	\$0.95
Claude Haiku 4.5	\$0.63	\$1.44	\$3.45
GPT-6.1 Sol	\$1.26	\$2.88	\$6.90
Claude Sonnet 5.5	\$1.26	\$2.88	\$6.90
Gemini 3.1 Pro	\$1.38	\$3.16	\$7.59
Claude Opus 5.5	\$2.53	\$5.77	\$13.80
GPT-6 Astra	\$6.32	\$14.42	\$34.50

Low scenario: 300K input + 50K output. Base: 654K + 120K. High: 1.5M + 300K. Each includes 15% overhead.

Reasoning models may bill hidden/thinking output according to provider rules, so real usage can exceed simple visible-text estimates.

7. Cohort-Level AI Budget

Estimated model-only semester spend under the base workload assumption:

Model	50 students	100 students	250 students	500 students	1000 students
GPT-6 Luna	\$7	\$14	\$36	\$72	\$144
Gemini 3.1 Flash-Lite	\$20	\$40	\$99	\$198	\$395
Claude Haiku 4.5	\$72	\$144	\$361	\$721	\$1,442
GPT-6.1 Sol	\$144	\$288	\$721	\$1,442	\$2,884
Claude Sonnet 5.5	\$144	\$288	\$721	\$1,442	\$2,884
GPT-6 Astra	\$721	\$1,442	\$3,605	\$7,210	\$14,421

7.1 Example: 250-student pilot

Architecture	Student AI cost / semester	Approx monthly over 3.5 months	Comment
All GPT-6 Luna	\$36	\$10	Cheapest; quality must be validated for protected tutoring
Hybrid: 65% Luna / 35% Sol	\$276	\$79	Recommended cost-quality architecture after evals
All GPT-6.1 Sol	\$721	\$206	Simple, consistent research condition; still inexpensive

For a 250-student pilot under the base token assumptions, running every interactive student turn on GPT-6.1 Sol is approximately \$721 for the semester. A 65/35 Luna/Sol routing mix is approximately \$276. These numbers exclude code-execution infrastructure, database/hosting, observability, taxes, and provider-specific tool charges.

8. Recommended Multi-Model Architecture

Request type	Default	Escalation	Reason
Protected Socratic chat	GPT-6.1 Sol	GPT-6 Astra only for internal	Research consistency + coding

		evals, not arbitrary student escalation	quality
Simple practice explanation	GPT-6 Luna	GPT-6.1 Sol	Cheap first pass
Practice question generation	GPT-6 Luna	GPT-6.1 Sol	Generation can be precomputed/cached
Interactive card JSON	GPT-6 Luna	None normally	Bounded structured output
Topic classification	GPT-6 Luna	None normally	Cheap classification
Misconception extraction	GPT-6 Luna	GPT-6.1 Sol if low confidence	Structured label extraction
Faculty assignment generation	GPT-6.1 Sol	GPT-6 Astra for rare high-value benchmark tests	Low volume; quality first
Rubric feedback suggestion	GPT-6.1 Sol	Faculty human review	Impact on grades
Analytics weekly brief	GPT-6 Luna	GPT-6.1 Sol	Input is already computed metrics
Policy output check	GPT-6 Luna	Rule-based + Sol adjudication	Defense-in-depth, future work

8.1 Why not route each turn dynamically by 'difficulty' in the research study?

Aggressive dynamic routing can reduce cost, but it introduces a confound: two students in the same experimental condition might receive materially different model capabilities. For research V1, keep the core protected tutor model stable for the cohort. Dynamic routing is better suited to non-core tasks or a later production phase after the primary study.

9. Provider Options, Sorted by Cost

9.1 Economy tier

Model	Input	Output	Base \$/student	Best Socra use	Main caveat
OpenAI GPT-6 Luna	\$0.10	\$0.50	\$0.14	Classification/cards/quiz generation	Focused, high-volume tasks; 1.05M context.
Cohere Command R	\$0.15	\$0.60	\$0.20	Classification/cards/quiz generation	Low-cost conversational/RAG model; 128K context.
Google Gemini 3.1 Flash-Lite	\$0.25	\$1.50	\$0.40	Classification/cards/quiz generation	High-volume simple processing; 1M context; preview.
Google Gemini 3 Flash	\$0.50	\$3.00	\$0.79	Fast tutoring + generation	Fast general model; 1M context; preview.

9.2 Balanced tier

Model	Input	Output	Base \$/student	Fit for protected tutor
Anthropic Claude Haiku 4.5	\$1.00	\$5.00	\$1.44	Candidate
OpenAI GPT-6.1 Sol	\$2.00	\$10.00	\$2.88	Strong candidate
Anthropic Claude Sonnet 5.5	\$2.00	\$10.00	\$2.88	Strong candidate
Google Gemini 3.1 Pro	\$2.00	\$12.00	\$3.16	Strong candidate
OpenAI GPT-5.6 Terra	\$2.00	\$12.00	\$3.16	Strong candidate

9.3 Premium/frontier tier

Model	Input	Output	Base \$/student	Recommended use
Anthropic Claude Opus 5.5	\$4.00	\$20.00	\$5.77	Offline eval / authoring benchmark
OpenAI GPT-5.6 Sol	\$4.00	\$20.00	\$5.77	Offline eval / authoring benchmark
OpenAI GPT-6 Astra	\$10.00	\$50.00	\$14.42	Hard benchmark / high-value faculty generation, not default chat

10. Model Decision Matrix for Socra

Option	Cost score	Coding/Reasoning	Research simplicity	Engineering simplicity	Recommendation
OpenAI-only: Luna + GPT-6.1 Sol	5/5	4.5/5	5/5	5/5	BEST V1
OpenAI-only: all GPT-6.1 Sol	4/5	4.5/5	5/5	5/5	BEST SIMPLE V1
Google: Flash-Lite + Pro	4.5/5	4/5	4/5	4/5	Benchmark alternative
Anthropic: Haiku + Sonnet	3.5/5	4.5/5	4/5	4/5	Benchmark alternative
Multi-provider live routing	4/5	5/5 potential	2.5/5	2/5	Do later
Frontier-only (Astra/Opus)	1/5	5/5	5/5	4/5	Overkill for default usage

10.1 Recommended decision

Pilot default: GPT-6.1 Sol for the protected student Socratic tutor and faculty high-value generation; GPT-6 Luna for structured/high-volume auxiliary AI; text-embedding-3-small for course retrieval. Keep the protected tutor model fixed during the study unless the protocol explicitly permits model changes.

Before committing, run the same Socra-specific evaluation set against GPT-6.1 Sol, Claude Sonnet 5.5, Gemini 3.1 Pro, and one economy model. The decision should be based on protected-answer leakage, usefulness, coding diagnosis, false refusals, latency, and cost, not generic benchmark scores alone.

11. Expected Cost by AI Function

Function	Illustrative volume / course semester	Default model	Approx token volume	Approx cost
Protected tutor, 250 students	250 × base protected usage	GPT-6.1 Sol	126M input / 21M output + overhead	\$531
Practice, 250 students	250 × base practice usage	GPT-6 Luna	37.5M input / 9M output + overhead	\$9
Faculty authoring	~150 generation calls	GPT-6.1 Sol	1.2M input / 0.4M output	\$6
Misconception extraction	~50K extraction calls	GPT-6 Luna	25M input / 5M output	\$5
Weekly faculty briefs	14 weekly briefs × 2 courses	GPT-6 Luna	0.5M input / 0.05M output	\$0.08
Course embeddings	5M source tokens	Embedding-3-small	5M embedding tokens	\$0.10

The function-level table intentionally overstates some auxiliary workloads to show scale. The key insight remains the same: classification, extraction, and embeddings are extremely inexpensive relative to complex student-facing reasoning. The dominant AI spend is controlled by which model handles protected conversation and how much context is resent on every turn.

12. Cost Optimization Levers

Lever	Potential impact	Implementation priority
Prompt/context caching	Large savings when assignment/course context repeats	P0
Conversation summarization	Reduces repeated historical tokens	P0
Cheap-model routing	5×–20× lower cost for bounded tasks	P0
Pre-generated practice bank	Avoids live generation on every attempt	P0/P1
Structured outputs	Shorter responses; fewer retry loops	P0
Output token caps	Directly limits expensive generation	P0

Batch generation	Often ~50% discount with some providers	P1
Cached course embeddings	Embed once, retrieve many times	P0
Model fallback only on failure	Avoids premium default cost	P1
Per-course budget alerts	Prevents runaway spend	P0

12.1 Context caching example

Suppose a protected chat turn contains 10,000 tokens of stable assignment/system/course context plus 2,000 tokens of changing conversation and code. With GPT-6.1 Sol, uncached 10K stable context costs roughly \$0.02 per request at the \$2/M input rate. If eligible cached input is \$0.10/M, the same 10K cached context is about \$0.001. Across 100,000 turns, that difference is roughly \$1,900 in input cost before accounting for cache writes and provider-specific rules.

13. AI Gateway Architecture

- One server-side AI gateway owns provider credentials and model selection.
- Every request includes an explicit mode: PROTECTED_ASSESSMENT, PRACTICE, POST_ASSESSMENT_REVIEW, FACULTY_AUTHORIZING, or FACULTY_ANALYTICS.
- Gateway attaches prompt version, assignment policy version, course/assignment identifiers, research condition, and trace ID.
- Structured tasks use JSON schema outputs.
- Usage records store input tokens, cached tokens, output/reasoning tokens where exposed, request cost, latency, provider, model, and status.
- No provider key is ever sent to the browser.
- Fallback behavior must never change a study condition silently.

13.1 Routing pseudo-policy

A practical V1 routing policy can be simple: student protected chat always uses the study's fixed protected model; faculty high-value authoring uses the balanced model; all deterministic classification/extraction/card tasks use the economy model; post-assessment and practice use economy by default with explicit escalation if evaluation shows a quality threshold is missed.

14. AI Metrics to Instrument from Day One

Metric	Unit	Why it matters
AI cost / active student	\$/student/week and semester	Unit economics
AI cost / assignment	\$/assignment	Find expensive activities
AI cost / successful intervention	\$/recovery	Connect spend to learning process
Input tokens / turn	tokens	Detect context bloat
Output tokens / turn	tokens	Control verbosity/cost
Cached-token share	%	Caching effectiveness
Model escalation rate	%	Router quality
Protected leakage rate	% eval cases	Core product safety
False refusal rate	% legitimate help cases	Student usability
AI latency p50/p95	seconds	UX
AI failure rate	% requests	Reliability
Guided recovery rate	%	Pedagogical outcome

15. Build vs. Buy

Socra should buy foundation-model intelligence via APIs and build its differentiated layers: educational context, protected-assessment policy, course/workspace integration, learner evidence, analytics, evaluation, and research instrumentation.

Build in Socra	Buy/use API
Assessment lifecycle and policy modes	Foundation model inference
Workspace/code context assembly	Embeddings
Secure code execution	Optional moderation/safety primitives
Learner evidence model	Cloud infrastructure
Misconception taxonomy	Model provider caching primitives
Faculty analytics	Managed database/vector storage if desired
Research telemetry/evaluation	Model benchmark capability

16. Phased AI Strategy

Phase 0 — Evaluation

- Build 100–300 representative Socra prompts across coding, conceptual help, jailbreak attempts, syntax errors, runtime errors, and ambiguous student questions.
- Run GPT-6.1 Sol, Claude Sonnet 5.5, Gemini 3.1 Pro, GPT-6 Luna, and optionally Claude Haiku 4.5.
- Score answer leakage, usefulness, correctness, latency, and cost.
- Select the protected model before the study cohort begins.

Phase 1 — Research pilot

- Keep protected tutor model stable.
- Use one provider where possible.
- Use cheap auxiliary model for non-core structured tasks.
- Log every model/prompt/policy version.
- Optimize context and caching before optimizing model switching.

Phase 2 — Post-research production

- Introduce model-router quality thresholds.
- Benchmark providers continuously.
- Add confidence-based escalation.
- Use batch generation for question banks and analytics preprocessing.
- Negotiate enterprise/university pricing once volume is known.

17. Recommended Planning Budget

Pilot size	Lean AI budget	Comfortable AI budget	Suggested ceiling
50 students	\$50–\$150	\$200–\$400	\$500
100 students	\$100–\$300	\$300–\$700	\$1,000
250 students	\$250–\$750	\$750–\$1,500	\$2,500
500 students	\$500–\$1,500	\$1,500–\$3,000	\$5,000

These are recommended planning envelopes rather than computed vendor quotes. They intentionally leave headroom for inefficient early prompts, evaluation traffic, faculty testing, retries, model changes, and

unexpected usage. Under the base model-only calculations above, the actual protected-tutor token spend can be far below these ceilings.

The larger financial risks for the pilot are likely engineering time, secure code execution, production reliability, and compliance/operations, not ordinary text-token spend.

18. Final Recommendation

- Use OpenAI as the production provider for Pilot V1 unless Socra-specific evals show a material quality disadvantage.
- Use GPT-6.1 Sol as the protected Socratic tutor baseline.
- Use GPT-6 Luna for high-volume structured AI: cards, tagging, misconception extraction, practice generation, lightweight explanations, and analytics narration.
- Use text-embedding-3-small for V1 course-resource retrieval unless retrieval evaluation shows a need for a stronger embedding model.
- Keep GPT-6 Astra, Claude Opus 5.5, and other frontier models out of the default student path. Use them for offline benchmarks or rare high-value faculty tasks.
- Do not optimize around tiny embedding or classification costs before solving context bloat and protected-tutor quality.
- Keep the student-facing protected model fixed during a research condition to preserve interpretability and reproducibility.
- Implement per-request cost logging and per-course budgets before opening the pilot to students.

18.1 The strategic AI moat

The moat should not be a particular model. Model prices and rankings change quickly. Socra's defensible layer is the educational system around the models: protected assessment policy, context-aware intervention, coding-workspace integration, longitudinal learning evidence, misconception analytics, faculty intelligence, and a research-backed understanding of how much AI help supports learning without replacing it.

19. Pricing Sources and Caveats

Prices were checked against public provider pages on October 6, 2026. Re-check before purchasing or publishing cost claims.

- [OpenAI API pricing](#)
- [OpenAI GPT-6 Luna model](#)
- [OpenAI GPT-6 model catalog / comparison](#)
- [OpenAI text-embedding-3-small](#)
- [Anthropic Claude Sonnet 5.5](#)
- [Anthropic Claude Opus](#)
- [Anthropic Claude Haiku](#)
- [Google Gemini Developer API pricing](#)
- [Google Gemini 3 developer guide](#)
- [Cohere Command R](#)
- [Instructure + OpenAI partnership](#)

Pricing caveats: provider prices can change; some models have separate long-context, regional, fast/priority, batch, cache-write, or data-residency pricing. Reasoning/thinking tokens may be billed as output. Gemini 3-

series entries referenced here include preview models. OpenAI GPT-5.6 Sol promotional pricing is time-limited according to current pricing materials. All cohort costs in this document are planning calculations based on explicitly stated token assumptions, not measured Socra usage.

Appendix A. Cost Formula

For a standard text request:

$$\text{Request cost} = (\text{input tokens} \div 1,000,000 \times \text{input rate}) + (\text{output tokens} \div 1,000,000 \times \text{output rate})$$

Semester cost per student in the base planning model:

$$\text{Base} = [(654,000 \div 1\text{M} \times \text{input rate}) + (120,000 \div 1\text{M} \times \text{output rate})] \times 1.15$$

If caching applies, split input into uncached and cached tokens and price each separately. If provider reasoning tokens are billed in output, include them in the output-token quantity.

Appendix B. Base Workload Price Table

Provider	Model	Low	Base	High	3K/500 request
OpenAI	GPT-6 Luna	\$0.06	\$0.14	\$0.35	\$0.00055
Cohere	Command R	\$0.09	\$0.20	\$0.47	\$0.00075
Google	Gemini 3.1 Flash-Lite	\$0.17	\$0.40	\$0.95	\$0.00150
Google	Gemini 3 Flash	\$0.35	\$0.79	\$1.90	\$0.00300
Anthropic	Claude Haiku 4.5	\$0.63	\$1.44	\$3.45	\$0.00550
OpenAI	GPT-6.1 Sol	\$1.26	\$2.88	\$6.90	\$0.01100
Anthropic	Claude Sonnet 5.5	\$1.26	\$2.88	\$6.90	\$0.01100
Google	Gemini 3.1 Pro	\$1.38	\$3.16	\$7.59	\$0.01200
OpenAI	GPT-5.6 Terra	\$1.38	\$3.16	\$7.59	\$0.01200
Anthropic	Claude Opus 5.5	\$2.53	\$5.77	\$13.80	\$0.02200
OpenAI	GPT-5.6 Sol	\$2.53	\$5.77	\$13.80	\$0.02200
OpenAI	GPT-6 Astra	\$6.32	\$14.42	\$34.50	\$0.05500